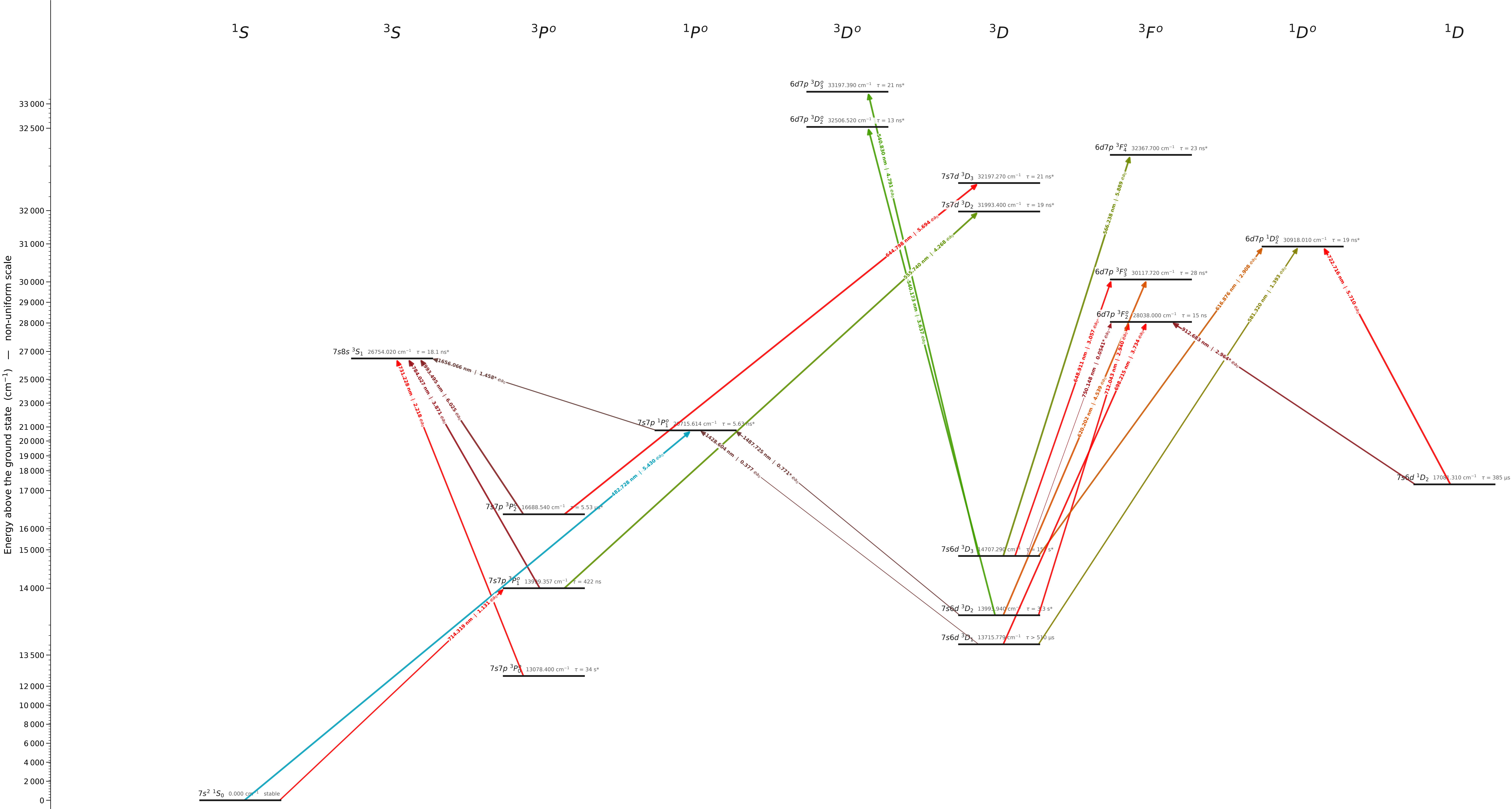
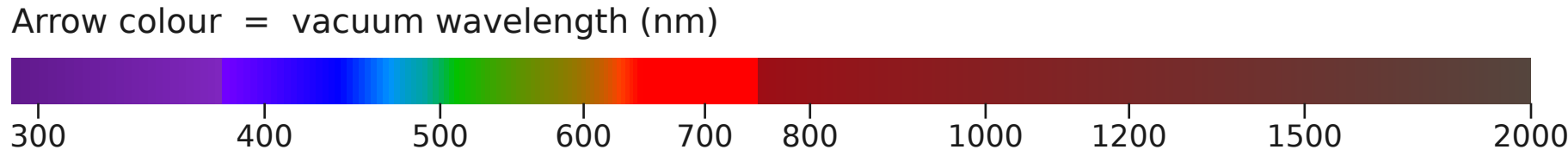


Radium

Ra (Ra I). Strongest classified transitions below 2 μm (NIST): 18 levels, 22 lines.

Ra⁺ ionisation limit 42573.36 cm⁻¹ = 5.27842 eV (234.889 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle U||er||U\rangle$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\langle U er U\rangle$ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
7s ² 1S ₀ - 7s7p 1P ₁ ^o	482.7277	482.5928	621.038489(15)	5.430	1.770e+08	28.27	99.7	strong singlet line
7s6d 3D ₃ - 6d7p 3F ₄ ^o	566.2383	566.0812	529.4458	5.889	4.300e+07	6.92	98.9	566.2 nm
7s6d 3D ₁ - 6d7p 3D ₂ ^o	581.3199	581.1588	515.7099	1.393	4.000e+06	8.377	7.6	581.3 nm
7s6d 3D ₃ - 6d7p 3D ₂ ^o	616.8757	616.7051	485.9852	2.908	1.460e+07	8.377	27.7	616.9 nm
7s6d 3D ₂ - 6d7p 3F ₃ ^o	620.2020	620.0304	483.3788	4.539	2.500e+07	5.684	70	620.2 nm
7s7p 3P ₂ ^o - 7s7d 3D ₃	644.7981	644.6200	464.9400	5.694	3.500e+07	7.579	73.5	644.8 nm line used for collinear laser spectroscopy at IS...
7s6d 3D ₃ - 6d7p 3F ₃ ^o	648.9112	648.7319	461.9931	3.057	9.900e+06	5.684	27.7	648.9 nm
7s6d 3D ₁ - 6d7p 3F ₂ ^o	698.2151	698.0226	429.369773(15)	3.734	1.660e+07	10.61	31	proposed spin-flip repump channel for the blue-slower sch...
7s6d 3D ₂ - 6d7p 3F ₂ ^o	712.0434	711.8472	421.031151(15)	2.940	9.700e+06	10.61	64	proposed spin-flip repump channel for the blue-slower sch...
7s ² 1S ₀ - 7s7p 3P ₁ ^o	714.3185	714.1217	419.69016(3)	1.131	2.370e+06	0.3771	100	intercombination line
7s6d 1D ₂ - 6d7p 1D ₂ ^o	722.7157	722.5165	414.8138	5.710	3.500e+07	8.377	66.5	722.7 nm
7s7p 3P ₀ ^o - 7s8s 3S ₁	731.2283	731.0269	409.9848	2.218	8.500e+06	8.793	15.2	731.2 nm
7s6d 3D ₃ - 6d7p 3F ₂ ^o	750.1476	749.9411	399.6446	0.054*	2.815e+03	10.61	0.4	750.1 nm
7s7p 3P ₁ ^o - 7s8s 3S ₁	784.0270	783.8113	382.3752	3.871	2.100e+07	8.793	38.3	784.0 nm
7s6d 1D ₂ - 6d7p 3F ₂ ^o	912.6828	912.4323	328.473887(15)	2.964*	4.683e+06	10.61	5	proposed spin-flip repump channel for the blue-slower sch...
7s7p 3P ₂ ^o - 7s8s 3S ₁	993.4946	993.2223	301.7555	6.025	2.500e+07	8.793	45.9	993.5 nm
7s6d 3D ₁ - 7s7p 1P ₁ ^o	1428.6041	1428.2136	209.8499(6)	0.377	3.300e+04	28.27	0.0231	repumper
7s7p 1P ₁ ^o - 7s8s 3S ₁	1656.0662	1655.6140	181.0268	1.458*	3.159e+05	8.793	0.572	1656 nm

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
7s ² 1S ₀ - 7s7p 1P ₁ ^o	482.728	208-214	8485(8)
7s ² 1S ₀ - 7s7p 1P ₁ ^o	482.728	209-214	8313(9)
7s ² 1S ₀ - 7s7p 1P ₁ ^o	482.728	210-214	6003(6)
7s ² 1S ₀ - 7s7p 1P ₁ ^o	482.728	211-214	5522(4)
7s ² 1S ₀ - 7s7p 1P ₁ ^o	482.728	212-214	3256(3)
7s ² 1S ₀ - 7s7p 1P ₁ ^o	482.728	213-214	2167(3)
7s ² 1S ₀ - 7s7p 1P ₁ ^o	482.728	220-214	-21904(8)
7s ² 1S ₀ - 7s7p 1P ₁ ^o	482.728	221-214	-25945(8)
7s ² 1S ₀ - 7s7p 1P ₁ ^o	482.728	222-214	-28827(8)
7s ² 1S ₀ - 7s7p 1P ₁ ^o	482.728	223-214	-32453(9)
7s ² 1S ₀ - 7s7p 1P ₁ ^o	482.728	224-214	-35123(9)
7s ² 1S ₀ - 7s7p 1P ₁ ^o	482.728	225-214	-38888(10)
7s ² 1S ₀ - 7s7p 1P ₁ ^o	482.728	226-214	-41124(11)
7s ² 1S ₀ - 7s7p 1P ₁ ^o	482.728	227-214	-43937(12)
7s ² 1S ₀ - 7s7p 1P ₁ ^o	482.728	228-214	-46950(12)
7s ² 1S ₀ - 7s7p 1P ₁ ^o	482.728	229-214	-50066(15)
7s ² 1S ₀ - 7s7p 1P ₁ ^o	482.728	230-214	-53634(15)
7s ² 1S ₀ - 7s7p 1P ₁ ^o	482.728	232-214	-59585(17)
7s ² 1S ₀ - 7s7p 1P ₁ ^o	482.728	225-226	2236(15)
7s7p 3P ₂ ^o - 7s7d 3D ₃	644.798	212-214	-887(4)
7s7p 3P ₂ ^o - 7s7d 3D ₃	644.798	221-214	7001(7)
7s7p 3P ₂ ^o - 7s7d 3D ₃	644.798	222-214	7781(4)

Shift = ν(first isotope) - ν(second isotope).

²²¹Ra hyperfine structure (*I* = 5/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
7s7p 3P ₁ ^o	-475.8(14)	-747.6(26)	F=3/2: +1141.98 F=5/2: +1073.88 F=7/2: -1376.40
7s7p 3P ₂ ^o	278.6(9)	1104(4)	F=1/2: -1177.54 F=3/2: -1256.35 F=5/2: -1111.75 F=7/2: -329.81 F=9/2: +1668.95
7s7p 1P ₁ ^o	136.3(8)	670(3)	F=3/2: -8.33 F=5/2: -671.98 F=7/2: +508.15
7s7d 3D ₃	213.3(6)	118(4)	F=1/2: -2062.32 F=3/2: -1774.18 F=5/2: -1282.16 F=7/2: -568.59 F=9/2: +391.26 F=11/2: +1629.20

Measured A, B constants; sources in the data file.